

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Industry Problem Solving

Code of Conduct:

I Moshitha kolanjiyappan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Moshitha k

Annexure A

Short Answer Test

1, A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.

- Store customer purchase history as cases/features such as products bought, frequency, and spending.
- Measure similarity using **Euclidean distance** or **cosine similarity**.
- KNN finds the most similar customers and recommends products they purchased.
- Case-Based Learning retrieves similar past customers and reuses their successful recommendations.
- New customer data is added to the database, so the system continuously adapts.

2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.

- Collect sensor values such as temperature, pressure, vibration, and speed.
- For a new input, assign **higher weights to nearby observations** and lower weights to distant ones.
- Build a local regression model using these weighted samples.
- Local models capture changing machine behavior better than one global model.
- Missing/noisy data can be handled using preprocessing, imputation, normalization, and outlier removal.

3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.

- Use features such as age, blood pressure, test results, and medical measurements.
- KNN classifies a patient based on the classes of its nearest patients.
- RBF networks use radial basis functions to capture complex relationships.
- Feature selection removes irrelevant attributes and improves accuracy.
- For large datasets, indexing, dimensionality reduction, and suitable values of **K** improve efficiency and prediction quality.

4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics

are defined, how past cases are reused, and how the system scales with increasing product data.

- Represent each product using price, category, brand, specifications, and customer ratings.
- Calculate similarity using **cosine similarity, Euclidean distance, or weighted feature similarity**.
- Retrieve similar historical products and reuse their pricing information.
- Adjust the price based on demand, competition, and previous successful cases.
- For scalability, use databases, clustering, indexing, and approximate nearest-neighbor search.

5. A logistics company needs to forecast delivery times based on location and traffic patterns. Propose a solution using Radial Basis Function networks or Locally Weighted Regression. Explain how the model captures non-linear relationships and improves prediction reliability under varying conditions.

- Inputs include distance, location, traffic, weather, time, and delivery type.
- **RBF networks** use radial basis functions to model non-linear relationships.
- Alternatively, LWR gives greater importance to observations from similar traffic/location conditions.
- The model learns different delivery patterns under different conditions.
- Continuous updates with recent traffic and delivery data improve reliability.

6. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using K-Nearest Neighbour or Logistic Regression. Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.

- Use features such as income, credit score, loan amount, repayment history, and transaction behavior.
- KNN compares a customer with similar historical customers.
- Logistic Regression creates a decision boundary and estimates default probability.
- Feature selection identifies the most important financial attributes.
- New repayment and transaction data can be periodically added to retrain/update the model.

7. A smart agriculture system aims to predict crop yield based on soil quality, weather conditions, and irrigation levels. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning improves predictions and how the system handles seasonal variations and noisy environmental data.

- Collect soil quality, rainfall, temperature, irrigation, fertilizer, and crop information.
- LWR gives higher weight to observations with similar environmental conditions.
- RBF networks can model non-linear relationships between environmental factors and yield.
- Seasonal information can be included as features such as month, rainfall period, and temperature.

- Missing values, sensor noise, and outliers are handled using preprocessing and robust data cleaning.

8. A cybersecurity firm wants to identify abnormal network behavior to detect intrusions. Design a system using KNN or Case-Based Learning. Explain how similarity between network patterns is measured, how anomalies are detected, and how the model updates with new attack patterns.

- Represent network activity using features such as packet rate, protocol, connection duration, and bytes transferred.
- Measure similarity using Euclidean or cosine distance.
- KNN compares new traffic with known normal and attack patterns.
- A new pattern that is significantly different from normal cases can be marked as an anomaly.
- Newly detected attack patterns are stored as cases, allowing the system to recognize future attacks.

9. A streaming platform aims to recommend movies to users based on viewing history and ratings. Propose a solution using KNN or Case-Based Learning. Explain how user similarity is calculated, how recommendations are generated, and how the system handles large-scale user data efficiently.

- Represent users using viewing history, ratings, genres, and watch frequency.
- Calculate user similarity using **cosine similarity** or correlation.
- KNN identifies users with similar preferences.
- Recommend movies highly rated or watched by similar users.
- For large-scale systems, use clustering, dimensionality reduction, and approximate nearest-neighbor techniques.

10. A transportation company wants to optimize route selection based on traffic, distance, and fuel consumption. Design a solution using RBF networks or Locally Weighted Regression. Explain how the model captures complex relationships between variables and improves decision-making in dynamic and real-time environments.

- Use traffic density, distance, fuel consumption, road conditions, and travel time as inputs.
- RBF networks learn complex non-linear relationships among these variables.
- LWR can focus on routes with conditions similar to the current situation.
- Predict travel time and fuel consumption for available routes.
- Real-time traffic updates allow the system to dynamically select the most efficient route.

11. A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time.

- Use data usage, call duration, SMS usage, recharge frequency, and current plan information.
- Measure similarity between users using normalized feature distance or cosine similarity.
- KNN identifies customers with similar usage patterns.
- Recommend plans that worked well for similar customers.
- Continuously update customer profiles with recent usage so recommendations adapt to changing behavior.

12. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

- Use time, temperature, weather, appliance usage, holidays, and previous consumption.
- LWR gives more importance to observations with similar time and environmental conditions.
- RBF networks model non-linear consumption patterns.
- Seasonal features help identify daily, weekly, and yearly variations.
- Noise can be reduced using smoothing/outlier removal, while missing readings can be handled using interpolation or imputation

13. A fraud detection system in an online payment platform needs to classify transactions as legitimate or fraudulent. Design a solution using KNN or Logistic Regression. Explain how similarity or decision boundaries are used, how features are selected, and how the model adapts to evolving fraud patterns.

- Extract features such as transaction amount, location, time, device, frequency, and transaction history.
- KNN compares a transaction with previous legitimate and fraudulent transactions.
- Logistic Regression uses a decision boundary to estimate the probability of fraud.
- Feature selection identifies important fraud indicators and reduces unnecessary computation.
- New confirmed fraud cases are added to the training data to adapt to evolving fraud patterns.

14. A manufacturing company wants to detect defective products based on sensor readings and quality measurements. Propose a solution using RBF networks or Case-Based Learning. Explain how non-linear relationships are captured, how similarity between cases is measured, and how the system improves with additional production data.

- Collect sensor readings, dimensions, temperature, pressure, and quality measurements.
- RBF networks capture non-linear relationships between manufacturing conditions and defects.
- Case-Based Learning compares a new product with previously classified defective/non-defective products.
- Similarity can be calculated using weighted Euclidean distance.
- As more production data becomes available, the system learns new defect patterns and improves classification.

15. A smart city system aims to predict traffic flow and congestion levels using data from sensors, weather, and time of day. Design a solution using Locally Weighted Regression

or KNN. Explain how the model captures dynamic patterns, how predictions are made in real-time, and how scalability is maintained with large volumes of data.

- Collect traffic sensor data, weather, time, road occupancy, and vehicle speed.
- KNN finds historical traffic situations similar to the current situation.
- LWR gives greater weight to recent and locally similar traffic conditions.
- Predictions can be generated in real time using the latest sensor readings.
- For scalability, use data sampling, clustering, indexing, and distributed processing for large sensor datasets.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand